

Ethereum blockchain. Creators can monetise voxel assets.

Second Life: Second Life's Metaverse City bills itself as "a welcoming role-play community" with immersive experiences that let players come in and out as they like. It's a relatively unsecured space.

Somnium Space: Open source platform Somnium Space is termed as more of a consumer gaming platform. Users can purchase digital land parcels and build homes and buildings, as well as start businesses, among other activities.

The risks

While the metaverse seems exciting and enjoyable, it has its own set of risks and drawbacks.

Harassment and cyberbullying: Young adults and teens are vulnerable to cyberbullying. Its impact, which can range from a feeling of low self-worth to suicidal impulses, particularly among teens, has been well-documented. According to experts, because the human experience in the metaverse is as genuine as in the physical world, pain and suffering too are as real and severe.

Some hazards are more difficult to prevent in a virtual environment. Many free-to-play games, for example, rely on advertisements to fund their growth. Malicious people may hypothetically replace these advertisements with images that cause sickness, which may be sent to a virtual reality headset.

Theft of identity: Many experts are afraid that if rigorous security measures are not adopted, identity theft in the metaverse may become even easier. In the real world, identity theft is now a multibillion-dollar industry, with damages being estimated at US\$ 24 billion.

Collection of unauthorised data by companies: Your personal information is today collected by legitimate businesses. Virtual reality, on the other hand, has the potential

to push data collection to a new level. Virtual reality headsets, for example, may hypothetically allow third parties to collect sensitive personal data including voiceprint data, biometric data, and even facial geometry data.

Ransomware attacks: Ransomware is malicious software that encrypts your data and prevents you or anyone else from accessing it. It will then display a message demanding you to pay a particular amount of money to regain access to your data, thus the term 'ransomware'. Your metaverse profile will have a lot more information than a regular social media page; it will also include a lot of sensitive information. Hence, such attacks will have a much larger impact.

A change in real world perceptions: According to a study done at Stanford University, both virtual reality and augmented reality, two of the foundations of metaverse, can influence how individuals see the actual world. Participants in that study, for example, avoided sitting in a chair where a computer-generated avatar had sat in their AR environment.

Deepfake videos: In a world where information is king, experts are concerned about bogus information campaigns delivered via deep falsified audio and video snippets, endangering a country's security. Deepfakes are modified video or audio clips that appear and/or sound like someone else.

Attacks using social engineering: The activity of mentally influencing someone into giving sensitive information is known as social engineering. Because of the large quantity of personal data that will be kept in the metaverse, it might become a gold mine for hackers trying to sell personal data on the Dark Web. Education will be the foundation for metaverse security management.

Shared spaces come with their own set of dangers: The metaverse is built on the idea of bringing people together. While this might be beneficial

in certain respects, it can also raise issues. In the metaverse, you will have to deal with people who have contrary viewpoints. People behave differently in a virtual environment than they do in the actual world, according to studies. This is particularly evident in the field of massively multiplayer online role-playing games (MMORPGs), where veteran players frequently criticise newcomers and even abuse girls.

New applications will need to be vetted: New apps, like today's internet, have the potential to disrupt our digital life, leading to digital risk. The enormous volume of sensitive data that will be retained in a metaverse scenario, however, can make the damage considerably worse. We'll need to come up with ways to ensure that all new apps are scanned for harmful code.

Key technologies integrated with the metaverse platform

Blockchain: Blockchain technology forms the bedrock of the metaverse, as most of the applications run on blockchains. It provides the decentralisation and transparency that is necessary for the functioning of the metaverse. Blockchain technology helps developers integrate several functions in the metaverse such as digital ownership verification, value transfer, governance, digital collectability, accessibility and interoperability.

Cryptocurrency: Cryptocurrency is one of the primary technologies used in the metaverse. The platforms on metaverse only accept cryptocurrencies. So, users have to change their real-world currencies to crypto before they can carry out any transaction in the metaverse. It can be used to purchase resources such as NFTs, digital real estate and to make in-game purchases. In the metaverse, cryptocurrencies have massive value. For example,